

Let $\text{row}(X)$ denote the number of rows of a matrix X . Let

$$A \in \mathbb{R}^{m \times d} \quad (1)$$

be the full data matrix. Let A_w denote the current row-window, let $A_{<w}$ denote all rows seen before the current window, and let $B \in \mathbb{R}^{r \times d}$ denote the sketch carried from the previous windows. Write

$$A_w \in \mathbb{R}^{k \times d}, \quad k = \text{row}(A_w), \quad k > 1, \quad (2)$$

$$t_w := (\# \text{ rows already seen up to and including the current window}), \quad (3)$$

$$v \in \mathbb{S}^{d-1}. \quad (4)$$

For any matrix X with a nonzero response on v , write

$$h_2 \left(\frac{Xv}{\|Xv\|_2} \right) = -\log \left(\frac{\|Xv\|_4^4}{\|Xv\|_2^4} \right). \quad (5)$$

Exact quantities we would like to use. The exact old-plus-new operator at window w is

$$N_w := \begin{pmatrix} A_{<w} \\ A_w \end{pmatrix}. \quad (6)$$

If the old rows were still available, the exact second moment, fourth moment, entropy, and score for a direction v would be

$$e_w^*(v) := \|N_w v\|_2^2, \quad (7)$$

$$q_w^*(v) := \|N_w v\|_4^4, \quad (8)$$

$$h_{2,w}^*(v) := h_2 \left(\frac{N_w v}{\|N_w v\|_2} \right) = -\log \frac{q_w^*(v)}{(e_w^*(v))^2}, \quad (9)$$

$$\text{score}_{N_w}(v) := \|N_w v\|_2 \left(\frac{t_w}{d} \right)^{-\frac{h_{2,w}^*(v)}{4 \log t_w}}. \quad (10)$$

The streamed objective is constructed to approximate $\text{score}_{N_w}(v)$ without storing $A_{<w}$.

Available carried summary. The previous windows are represented by

$$B = SV^\top, \quad V = [v_1, \dots, v_r], \quad S = \text{diag}(s_1, \dots, s_r), \quad (11)$$

together with carried entropy values $h_1, \dots, h_r$. For a candidate direction v , define

$$a = V^\top v, \quad a_j = v_j^\top v. \quad (12)$$

The compact streamed operator used by the algorithm is

$$M_w := \begin{pmatrix} B \\ A_w \end{pmatrix}. \quad (13)$$

Streamed moment and entropy surrogates. The carried summary gives the following approximations to the exact quantities $e_w^*(v)$ and $q_w^*(v)$:

$$e_w(v) := \sum_{j=1}^r s_j^2 (v_j^\top v)^2 + \|A_w v\|_2^2, \quad (14)$$

$$q_w(v) := \sum_{j=1}^r s_j^4 e^{-h_j} (v_j^\top v)^4 + \|A_w v\|_4^4. \quad (15)$$

Since $B = SV^\top$, the second-moment surrogate is exactly the squared norm of the compact streamed operator:

$$e_w(v) = \|Bv\|_2^2 + \|A_w v\|_2^2 = \|M_w v\|_2^2. \quad (16)$$

The fourth-moment surrogate $q_w(v)$ models the old fourth moment using the stored per-direction entropies and adds the current-window fourth moment exactly. The streamed entropy surrogate is

$$\hat{h}_{2,w}(v) := -\log \frac{q_w(v)}{e_w(v)^2}. \quad (17)$$

Why $\hat{h}_{2,w}$ approximates the exact entropy. Let

$$y_{\text{old}} := A_{<w} v, \quad y_{\text{new}} := A_w v. \quad (18)$$

The stacked old and new responses occupy disjoint row coordinates, so

$$\|N_w v\|_4^4 = \|y_{\text{old}}\|_4^4 + \|y_{\text{new}}\|_4^4, \quad \|N_w v\|_2^2 = \|y_{\text{old}}\|_2^2 + \|y_{\text{new}}\|_2^2. \quad (19)$$

To state the approximation error, decompose the old response into the retained coordinates and a residual component with coefficient $a_\perp$, gain $s_\perp$, and entropy $h_\perp$. Define

$$R_{4,w} := a_\perp^4 s_\perp^4 e^{-h_\perp} + R_{4,w}^{\text{mix}}, \quad (20)$$

$$E_{2,w} := a_\perp^2 s_\perp^2 + E_{2,w}^{\text{mix}}, \quad (21)$$

where

$$R_{4,w}^{\text{mix}} := \|y_{\text{old}}\|_4^4 - \sum_{j=1}^r a_j^4 s_j^4 e^{-h_j} - a_\perp^4 s_\perp^4 e^{-h_\perp}, \quad (22)$$

$$E_{2,w}^{\text{mix}} := \|y_{\text{old}}\|_2^2 - \sum_{j=1}^r a_j^2 s_j^2 - a_\perp^2 s_\perp^2. \quad (23)$$

Then the exact entropy ratio can be written as

$$\frac{\|N_w v\|_4^4}{\|N_w v\|_2^4} = \frac{q_w(v) + R_{4,w}}{(e_w(v) + E_{2,w})^2}. \quad (24)$$

Consequently, if

$$|R_{4,w}| \ll e_w(v)^2, \quad |E_{2,w}| \ll e_w(v), \quad (25)$$

then

$$h_{2,w}^*(v) = \hat{h}_{2,w}(v) + \mathcal{O}\left(\frac{|R_{4,w}|}{e_w(v)^2} + \frac{|E_{2,w}|}{e_w(v)} + \frac{E_{2,w}^2}{e_w(v)^2}\right). \quad (26)$$

Thus $\hat{h}_{2,w}$ approximates the exact entropy $h_2(N_w v / \|N_w v\|_2)$ when the omitted old residual and old mixing terms are small relative to the retained second-moment scale.

Streamed score. The exact score $\text{score}_{N_w}(v)$ depends on both $\|N_w v\|_2$ and $h_{2,w}^*(v)$. The sketch is designed so that

$$\|N_w v\|_2 \approx \|M_w v\|_2, \quad (27)$$

while the previous paragraph gives

$$h_{2,w}^*(v) \approx \hat{h}_{2,w}(v). \quad (28)$$

Substituting these approximations into the exact score motivates the streamed score

$$\widehat{\text{score}}_w(v) := \|M_w v\|_2 \left(\frac{t_w}{d}\right)^{-\frac{\hat{h}_{2,w}(v)}{4 \log t_w}}. \quad (29)$$

This is the sense in which the later-window objective approximates the exact score of the unavailable operator N_w .

First-window special case. For the first window, there is no previous sketch:

$$B = 0, \quad M_1 = N_1 = A_1. \quad (30)$$

Therefore

$$e_1(v) = \|A_1 v\|_2^2, \quad q_1(v) = \|A_1 v\|_4^4, \quad \hat{h}_{2,1}(v) = h_2\left(\frac{A_1 v}{\|A_1 v\|_2}\right), \quad (31)$$

so the entropy and score are exact for the first observed window.

Algorithm overview. The method processes the matrix one row-window at a time.

For the first window, define

$$\text{score}_1(v) := \|A_1 v\|_2 \left(\frac{\text{row}(A_1)}{d}\right)^{-\frac{h_2\left(\frac{A_1 v}{\|A_1 v\|_2}\right)}{4 \log \text{row}(A_1)}}. \quad (32)$$

The retained directions are computed sequentially by solving

$$v_j \in \arg \max_{v \in \mathbb{S}^{d-1}, v_i^\top v = 0 \ \forall i < j} \text{score}_1(v). \quad (33)$$

After the selected directions are accepted, the sketch carried to the next window is

$$V_1 = [v_1, \dots, v_r], \quad (34)$$

$$s_j^{(1)} := \|M_1 v_j\|_2, \quad h_j^{(1)} := \widehat{h}_{2,1}(v_j), \quad j = 1, \dots, r, \quad (35)$$

$$S_1 = \text{diag}(s_1^{(1)}, \dots, s_r^{(1)}), \quad h^{(1)} = (h_1^{(1)}, \dots, h_r^{(1)}), \quad (36)$$

$$B^{(1)} = S_1 V_1^\top. \quad (37)$$

For a later window w , the method receives the previous sketch

$$V_{w-1} = [v_1^{(w-1)}, \dots, v_r^{(w-1)}], \quad (38)$$

$$S_{w-1} = \text{diag}(s_1^{(w-1)}, \dots, s_r^{(w-1)}), \quad (39)$$

$$h^{(w-1)} = (h_1^{(w-1)}, \dots, h_r^{(w-1)}), \quad (40)$$

$$B = B^{(w-1)} = S_{w-1} V_{w-1}^\top, \quad (41)$$

where

$$s_j^{(w-1)} := \|M_{w-1} v_j^{(w-1)}\|_2, \quad (42)$$

$$h_j^{(w-1)} := \widehat{h}_{2,w-1}(v_j^{(w-1)}), \quad j = 1, \dots, r. \quad (43)$$

It forms M_w , $e_w(v)$, $q_w(v)$, $\widehat{h}_{2,w}(v)$, and $\widehat{\text{score}}_w(v)$ as defined above. The retained directions are computed sequentially from

$$v_j^{(w)} \in \arg \max_{v \in \mathbb{S}^{d-1}, (v_i^{(w)})^\top v = 0 \ \forall i < j} \widehat{\text{score}}_w(v). \quad (44)$$

After solving for the updated directions, the next sketch is again the direct singular-factor summary

$$V_w = [v_1^{(w)}, \dots, v_r^{(w)}], \quad (45)$$

$$s_j^{(w)} := \|M_w v_j^{(w)}\|_2, \quad h_j^{(w)} := \widehat{h}_{2,w}(v_j^{(w)}), \quad j = 1, \dots, r, \quad (46)$$

$$S_w = \text{diag}(s_1^{(w)}, \dots, s_r^{(w)}), \quad h^{(w)} = (h_1^{(w)}, \dots, h_r^{(w)}), \quad (47)$$

$$B^{(w)} = S_w V_w^\top. \quad (48)$$

Derivation of the entropy correction factor. Let

$$z = Xv \quad (49)$$

be a row-restricted response and let

$$x = Av \tag{50}$$

be the response of the full matrix. We use the normalized-entropy heuristic

$$\frac{h_2\left(\frac{x}{\|x\|_2}\right)}{\log d} \approx \frac{h_2\left(\frac{z}{\|z\|_2}\right)}{\log \text{row}(X)}, \tag{51}$$

that is,

$$h_2\left(\frac{x}{\|x\|_2}\right) \approx h_2\left(\frac{z}{\|z\|_2}\right) \frac{\log d}{\log \text{row}(X)}. \tag{52}$$

Using

$$h_2\left(\frac{y}{\|y\|_2}\right) = 4 \log \|y\|_2 - 4 \log \|y\|_4, \tag{53}$$

and $\|z\|_4 \leq \|x\|_4$ for a restriction z of x , we obtain

$$4 \log \frac{\|z\|_2}{\|x\|_2} \leq h_2\left(\frac{z}{\|z\|_2}\right) - h_2\left(\frac{x}{\|x\|_2}\right). \tag{54}$$

Substituting the heuristic entropy scaling gives

$$\log \frac{\|z\|_2}{\|x\|_2} \lesssim -\frac{h_2\left(\frac{z}{\|z\|_2}\right)}{4} \frac{\log(d/\text{row}(X))}{\log \text{row}(X)}, \tag{55}$$

hence

$$\frac{\|Xv\|_2}{\|Av\|_2} \lesssim \left(\frac{\text{row}(X)}{d}\right)^{\frac{h_2\left(\frac{Xv}{\|Xv\|_2}\right)}{4 \log \text{row}(X)}}. \tag{56}$$

This motivates the reciprocal entropy factor

$$\left(\frac{\text{row}(X)}{d}\right)^{-\frac{h_2\left(\frac{Xv}{\|Xv\|_2}\right)}{4 \log \text{row}(X)}}. \tag{57}$$

Thus, for an exact window matrix X , the entropy-corrected score is

$$\text{score}_X(v) := \|Xv\|_2 \left(\frac{\text{row}(X)}{d}\right)^{-\frac{h_2\left(\frac{Xv}{\|Xv\|_2}\right)}{4 \log \text{row}(X)}}. \tag{58}$$

Simplified log form. Let

$$\gamma_w := \frac{\log(t_w/d)}{2 \log t_w}. \quad (59)$$

Then

$$\log \widehat{\text{score}}_w(v) = \frac{1}{2} \log \|M_w v\|_2^2 + \gamma_w \log q_w(v) - 2\gamma_w \log e_w(v) \quad (60)$$

$$= \frac{1}{2} \log e_w(v) + \gamma_w \log q_w(v) - 2\gamma_w \log e_w(v). \quad (61)$$

Equivalently, with

$$c_w := \frac{2 \log(t_w/d)}{\log t_w}, \quad (62)$$

we have

$$f_w(v) := \log \widehat{\text{score}}_w(v) = \frac{1 - c_w}{2} \log e_w(v) + \frac{c_w}{4} \log q_w(v). \quad (63)$$

This is the same interpolation between quadratic energy and fourth-moment concentration as in the first window, but now with the old moments represented by the carried sketch.

Gradient. For the first window,

$$\nabla f_1(v) = (1 - c_1) \frac{A_1^\top A_1 v}{\|A_1 v\|_2^2} + c_1 A_1^\top \left(\frac{(A_1 v)^{\odot 3}}{\|A_1 v\|_4^4} \right), \quad (64)$$

where

$$c_1 := \frac{\log(\text{row}(A_1)/d)}{\log \text{row}(A_1)}. \quad (65)$$

For later windows, differentiate the streamed quantities:

$$\nabla e_w(v) = 2V((s^{\odot 2}) \odot a) + 2A_w^\top A_w v, \quad (66)$$

$$\nabla q_w(v) = 4V((s^{\odot 4} \odot e^{-h}) \odot a^{\odot 3}) + 4A_w^\top ((A_w v)^{\odot 3}). \quad (67)$$

Therefore

$$\nabla f_w(v) = \frac{1}{2} \frac{\nabla \|M_w v\|_2^2}{\|M_w v\|_2^2} + \gamma_w \frac{\nabla q_w(v)}{q_w(v)} - 2\gamma_w \frac{\nabla e_w(v)}{e_w(v)}. \quad (68)$$

Using $\nabla \|M_w v\|_2^2 = 2M_w^\top M_w v$, this is

$$\nabla f_w(v) = \frac{M_w^\top M_w v}{\|M_w v\|_2^2} + \gamma_w \frac{\nabla q_w(v)}{q_w(v)} - 2\gamma_w \frac{\nabla e_w(v)}{e_w(v)}. \quad (69)$$

This is the exact gradient of the streamed surrogate objective.

Stored-basis form. If

$$q_j := s_j^4 e^{-h_j}, \quad (70)$$

then

$$q_w(v) = \sum_{j=1}^r q_j (v_j^\top v)^4 + \|A_w v\|_4^4. \quad (71)$$

So the carried state needed by the next window is naturally

$$(V, s, h) \quad \text{or equivalently} \quad (V, s^{\odot 2}, q), \quad (72)$$

with

$$q = s^{\odot 4} \odot e^{-h}. \quad (73)$$

This is the low-rank moment summary used in the streamed objective.

Optimization problem. To extract a direction, solve

$$\max_{v \in \mathbb{S}^{d-1}, Q^\top v = 0} \widehat{\text{score}}_w(v), \quad (74)$$

where the columns of Q are the previously accepted directions.

Projected full-gradient ascent. Use the exact gradient of the streamed surrogate score and project it onto the feasible tangent space:

$$g_{\text{tan}}(v) := (I - QQ^\top - vv^\top) \nabla f_w(v). \quad (75)$$

Then take

$$\tilde{v}^+ = v + \eta g_{\text{tan}}(v), \quad v^+ = \frac{(I - QQ^\top) \tilde{v}^+}{\|(I - QQ^\top) \tilde{v}^+\|_2}, \quad (76)$$

with $\eta > 0$ chosen by backtracking line search on the streamed surrogate objective.